

Second and Higher Derivatives

Repeated power-rule differentiation, second derivatives through the product and chain rules, implicit second derivatives, and acceleration and concavity

The derivative ladder. A higher-order derivative is just differentiation done again — each rung is the derivative of the rung above it, and each rung has its own meaning. Simplify before climbing to the next rung; an unsimplified rung doubles the work on the one after it.

Rung	Notation	What it measures
function	$f, y, s(t)$	value, position
first	$f', \frac{dy}{dx}, v(t)$	rate of change, slope, velocity
second	$f'', \frac{d^2y}{dx^2}, a(t)$	concavity, acceleration
third and up	$f''', f^{(4)}$	rate of change of the rung above

Notation used throughout this sheet. No values shown: use the function given in each problem.

Directions.

- Show each rung of the ladder, not just the last one — write f' before f'' .
- Simplify each derivative before differentiating again.
- Give exact values (no decimal approximations) unless a problem asks otherwise, and state units where the problem gives them.

1. Find $f''(x)$ for $f(x) = x^5 - 3x^3 + 2x$.

Answer: _____

2. Find the third derivative $g'''(x)$ for $g(x) = 2x^4 - x^2 + 7$.

Answer: _____

3. Find $f''(x)$ for $f(x) = \sin(3x)$. The chain rule is needed on both rungs.

Answer: _____

4. Find $h''(x)$ for $h(x) = x^2e^x$, and write your answer with e^x factored out.

Answer: _____

5. A drone rises along a vertical track. Its height above the launch pad, in metres, is $s(t) = t^3 - 6t^2 + 9t$, where t is in seconds and $0 \leq t \leq 5$.

- (a) Find the acceleration function $a(t) = s''(t)$, in metres per second squared.
- (b) Find the time t , in seconds, at which the acceleration is zero.

Write $a(t)$ and then the time on the answer line.

Answer: _____

6. Let $f(x) = \ln x + x^3$ for $x > 0$.

- (a) Find $f''(x)$.
- (b) Evaluate $f''(2)$ and state what its sign says about the concavity of the graph of f at $x = 2$.

Write $f''(x)$ and then the value of $f''(2)$ on the answer line.

Answer: _____

- 7.** The circle $x^2 + y^2 = 25$ defines y implicitly as a function of x on the upper arc.
- (a) Differentiate implicitly to find $\frac{dy}{dx}$, then differentiate again to find $\frac{d^2y}{dx^2}$ in terms of x and y . Substitute the original equation to write it in terms of y alone.
- (b) Evaluate $\frac{d^2y}{dx^2}$ at the point $(3, 4)$.

Write the second derivative and then its value at $(3, 4)$ on the answer line.

Answer: _____

8. A particle moves along a line with position $s(t) = t^4 - 4t^3 + 12$ metres at time t seconds, $t \geq 0$.

- (a) Find the acceleration $a(t) = s''(t)$, in metres per second squared.
- (b) Find every time $t \geq 0$ at which the acceleration is zero.
- (c) Using your answer to (b), say on which interval the position graph is concave down.

Write $a(t)$ and then the times on the answer line.

Answer: _____